

Deklen Crocker

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EDUCATION

BSc. in Applied Science of Electrical Engineering

Edmonton, AB | Nov 2022

UNIVERSITY OF ALBERTA

Coursework: Digital Circuit Design, IC Design, Digital Logic,

WORK EXPERIENCE

JONES MICROWAVE INC. | EMBEDDED SYSTEM DEVELOPER

Edmonton, AB | Sept 2022 – Present

- Designed and implemented a dashboard using **React** and **TypeScript** to visualize data stored in **DynamoDB**, decreasing time to understand delivery driver work sessions by over 10 times.
- Ensured only authorized employees have access to the application by creating an **AWS Lambda@Edge** function to intercept and sign valid requests.
- Devised and deployed the infrastructure in TypeScript through **AWS CDK**, then created efficient algorithms to process data from a custom **REST API**, so it could scale to handle millions of deliveries.
- Setup a CI/CD pipeline and exceeded requirements in each stage, which lead to the application being pushed to production.

GOVT. OF THE NORTHWEST TERRITORIES | PROJECTS INTERN

Inuvik, NT | Apr 2021 - Aug 2021

- Developed a Personal Health Record (PHR) system in **Spring** based on the theoretical model outlined in [Mitu Kumar's thesis](#).
- Applied the mCP-ABE encryption scheme using the JPBC library, so patients have fine-grained access control over their health records with the ability to instantly revoke access.

PROJECTS

AUTOMATED SPEAR-PHISHER

PYTHON, PYSPARK, SELENIUM, APACHE, NLP, BIG DATA

A security research tool to send targeted spam messages on either Facebook or Twitter. The program analyzes the feed of its target to increase the effectiveness of the messages.

IMAGE REPOSITORY

JAVA, PHP, REACT, TYPESCRIPT, MAVEN, GCP, MYSQL

A full-stack image repository, where users can store their files. Created a GCP cloud function to implement CP-ABE encryption and used it to ensure files are secure at rest.

SKILLS

Languages: Java, C++, Python, C#, PHP, Prolog, Bash, C, Racket, SQL

Web Development: React, JavaScript, TypeScript, HTML/CSS

Technology: Git, AWS, GCP, Azure, Docker, Unity, Apache, $\LaTeX$, MongoDB, DynamoDB, Neo4j